SEAGLASS FOUNDRY

Forged For Speed. Refined by Craft.

Seaglass Globe™

A WGS-84 Ellipsoid, in Double Precision, All the Way Down

From orbit to a rooftop in one move — no cut, no scene swap, the same globe the whole way down. A precision virtual globe rendered on the GPU, with real elevation above and below the waterline, real imagery, geodesy that names its algorithm, your own data on the stack, and the Forge suite's GPU compute running where you are looking. MIL-STD-2525C military symbology and scenarios replayed on a timeline. Release 1.2.6 is about being right: every measurement on the WGS-84 ellipsoid, checked against independent references. The suite's optional 3D view — and the surface the agent draws on.

Orbit → rooftop

in one continuous move

precision is held relative to the camera, so the ground stays steady at street level, exactly where a single-precision engine starts to shimmer

2525C

military symbology, in the box

units with their amplifiers and tactical graphics, available from the first launch — nothing to download

WGS-84

real geodesy, method named

Karney's geodesic method for distance, azimuth and area, on the ellipsoid — named beside every measurement

What it is

Seaglass Globe is a precision virtual globe: a WGS-84 ellipsoid held in double precision and rendered on the GPU, with real elevation, real imagery, and geodesy that names its algorithm. It reads GeoJSON, KML, shapefiles and GeoPackage, any XYZ, WMS, WMTS or MVT service you can name — and it is the Forge suite's optional 3D view: install it with the Seaglass Control Center™ and it starts alongside the engines.

Why it matters

The engines are headless, and ForgeGIS Studio™ is where a person operates them on a flat map — but some answers want the Earth. The Globe has its own Assistant, talking to ForgeMind™: ask there and the contours, elevation profiles, draped rasters and scene entities are drawn on the globe instead of only described. The window you type into is the map that answers.

How it's built differently

The numbers on screen are the render's own: precision held relative to the camera so a rooftop does not shimmer, and a measurement that prints its method. It runs ForgeGIS™ from a node-graph editor docked under the map — drag a value and only the downstream steps re-run — and it can be flown by software over the Model Context Protocol.

New in 1.2.6 — a release about being right. Every measurement now runs on the WGS-84 ellipsoid, each line type by its own model: geodesics by Karney's method, the number QGIS and pyproj give, antipodes included; rhumb lines as the ellipsoidal loxodrome; circle, square, ellipse and quad areas on the ellipsoid rather than planar formulas. MGRS truncates as MGRS does, and UTM and MGRS are checked against PROJ and GEOTRANS to a centimetre. A DEM in US survey feet reads in its own units, never as metres. And the local endpoint now refuses a DNS-rebinding page or a cross-site request, with or without a token.

CONTINUED ON REVERSE →

WHO SEAGLASS GLOBE IS FOR

Forge suite evaluators and operators

Run the whole suite on one Windows x64 workstation with a real GPU and watch the agent draw its answers on the globe. Draw a site by hand and it is the same kind of scene entity the agent places — name it, and refer to it in the next question.

Programs that plan and review exercises

Units draw as MIL-STD-2525C symbols with their amplifiers, alongside phase lines, objectives, axes of advance and range fans. A scenario plays on a timeline — real time or compressed, forward or back, event to event — and the suite includes a brigade-advance demonstration with notional positions to try it on. The opt-in ForgeRelay™ recorder captures live DIS and Cursor-on-Target traffic in the same scenario format.

Analysts with their own data and their own services

A map is a stack, and the stack is yours: drag one layer over another, pull the opacity down, hide it and it is still there. Bring GeoJSON, KML, shapefiles — loose or zipped — or GeoPackage; style each file with its own colours, labels and a legend, and open an attribute table you can search, sort and export. Point it at a standards server and it asks what that server has, then draws the layer you pick.

A REAL COORDINATE SYSTEM, NOT A PICTURE OF ONE

New York to Los Angeles bows north, because on an ellipsoid that is the short way. Two clicks on open water return distance, forward and back azimuth — and the name of the algorithm that computed them, because a number without its method is not a measurement. The same point reads as DD, DMS, UTM and MGRS, and `Go to` takes a coordinate typed in decimal degrees or DMS; measure in metric, US or nautical units, and save a measurement as GeoJSON with its algorithm recorded in the file.

> **Compute where you are looking.** Point at the ground and ask what elevation data exists there. Type a contour interval and the labelled lines are the real elevations. Drag the sun's altitude and the hillshade re-computes and re-drapes while you are still holding the slider. Take an edge off the workflow and feed one output into two branches, and it runs as a graph.

> **In the suite.** Installed by the Seaglass Control Center™ alongside the engines, the Globe serves its MCP endpoint on port 7900, loopback only and bearer-token protected, and runs its own copy of the GIS engines — the suite recommends 8 GB of RAM with it installed. It needs a real GPU; an RDP session usually can't drive it. The release film also shows live feeds and a satellite catalogue; those are showcase builds, not part of the installed suite.

Evaluate the Forge suite — with the Earth in the room

The Technical Brief covers symbology and scenarios, geodesy and precision, rendering, data, the MCP surface, the compute and twin bridges, and deployment. Seaglass Globe is an opt-in component of the Forge suite, installed by the Seaglass Control Center, and is not sold separately. The Forge suite is commercially licensed; evaluation builds are available to qualified teams.

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